

COURSE-0000 — Institution

Course Title

Lecture Notes

Your Name

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Using this template

Lecture	Lesson 1 (Introduction)
Textbook	H&P [1] §1.1–1.3
Keywords	sidenotes, index, bilingual build

This chapter is a showcase of every macro the class provides. Delete it when you start a real course, but keep `02-lesson-skeleton.tex` as a starting point for new lessons.

1.1 Margin notes

The right-hand column is for anything that supports the main text without interrupting it. Numbered remarks use `\sidenote`¹ and unnumbered remarks use `\margin`.

Three labelled variants make the *kind* of side content visible at a glance:

- `\source{...}` – where the material comes from (lecture, textbook section, paper);
- `\tip{...}` – practical advice;
- `\caution{...}` – pitfalls and misconceptions.

1.2 Terms and the index

A *pipeline* is introduced with `\term`, which emphasizes the word, adds it to the index, and optionally prints a gloss in the margin (here: the Japanese term). Later mentions of a *pipeline* use `\term*` so the index does not fill up with repeated page numbers. Terms can be nested: *speedup* and *Amdahl's law* both go to the index.

Definition 1.1 (*Speedup*). For a change that makes a fraction f of the execution time s times faster, the overall speedup is

$$S = \frac{1}{(1-f) + f/s}. \quad (1.1)$$

Example 1.2. If $f = 0.9$ and $s \rightarrow \infty$, Equation (1.1) gives $S = 10$: the untouched 10 % bounds the achievable speedup.

Note 1.3. `definition`, `example` and `note` share one counter per chapter, so they can be cross-referenced with `\cref`.

Key point – Amdahl's law

Optimize the common case; the part you do not touch limits the gain. Margin macros work inside this box too.

1.3 Figures, tables and code

Small figures go in the margin (Figure 1.1); normal figures use the standard `figure` environment; wide figures use `widefigure`, which spans the body and the margin column (Figure 1.3). Figures shared between the two editions live in `figures/` and switch their labels with `\iflangja`.

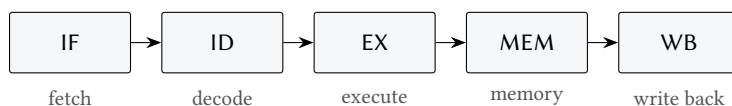


Figure 1.2: A classic five-stage pipeline (shared TikZ source).

¹ Like this one. Numbers restart at every chapter.

An unnumbered margin note.

Source Amdahl's original paper [2]; H&P §1.9.

Tip Use `\tip` for exam-relevant advice or mnemonics.

Caution Use `\caution` for common misconceptions.

▷ パイプライン

▷ Amdahl の法則

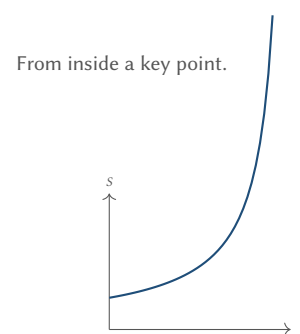


Figure 1.1: A figure in the margin (`marginfigure`).

Cycle	1	2	3	4	5	6	7
add	IF	ID	EX	MEM	WB		
sub		IF	ID	EX	MEM	WB	
lw			IF	ID	EX	MEM	WB

Figure 1.3: A wide float spanning body and margin (widefigure).

Code uses listings:

Listing 1.1: A loop in assembly.

```

1 | loop:   add    r1, r1, r2      ; accumulate
2 |         addi   r3, r3, -1
3 |         bne    r3, r0, loop    ; branch if not done

```

Pseudocode uses algorithmicx:

Algorithm 1 Two-bit saturating counter update

```

1: if taken then
2:    $c \leftarrow \min(c + 1, 3)$ 
3: else
4:    $c \leftarrow \max(c - 1, 0)$ 
5: end if

```

1.4 Bilingual writing

Both editions share `format/`, `figures/` and `references.bib`; only the prose under `en/` and `ja/` differs. Inside shared files use `\iflangja{japanese}{english}` or the shorthands `\ja{...}` and `\en{...}`.

Lesson skeleton

Lecture

Lesson N

2.1 First topic

Body text.

Key point

The one thing to remember from this section.

Source Lecture N, video 3.

2.2 Second topic

Body text.

Bibliography

- [1] John L. Hennessy and David A. Patterson. *Computer Architecture: A Quantitative Approach*. 6th ed. Morgan Kaufmann, 2017.
- [2] Gene M. Amdahl. “Validity of the Single Processor Approach to Achieving Large Scale Computing Capabilities”. In: *AFIPS Conference Proceedings* 30 (1967), pp. 483–485.

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